Signals and Systems Lecture-6: Linear Systems

Chadi Abou-Rjeily

Department of Electrical and Computer Engineering
Lebanese American University
chadi.abourjeily@lau.edu.lb

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Convolution (1)

Consider two continuous-time functions $x(t)$ and $h(t)$. The convolution between these functions is defined as:

$$y(t) = x(t) * h(t) = \int_{-\infty}^{+\infty} x(\tau)h(t-\tau)\mathrm{d}\tau$$

Note that:

- The convolution of two time functions is a time function.
- τ is a dummy variable: the result of the convolution will not depend on the value of τ (which will be integrated out).

Convolution (2)

Proposition:

$$x(t) * h(t) = h(t) * x(t)$$

Proof:

$$x(t) * h(t) = \int_{-\infty}^{+\infty} x(\tau)h(t-\tau)\mathrm{d}\tau$$

- Set $\tau' = t - \tau \Rightarrow \tau = t - \tau'$
  - for $\tau \to -\infty$, $\tau' \to +\infty$
  - for $\tau \to +\infty$, $\tau' \to -\infty$
  - $\mathrm{d}\tau' = -\mathrm{d}\tau$

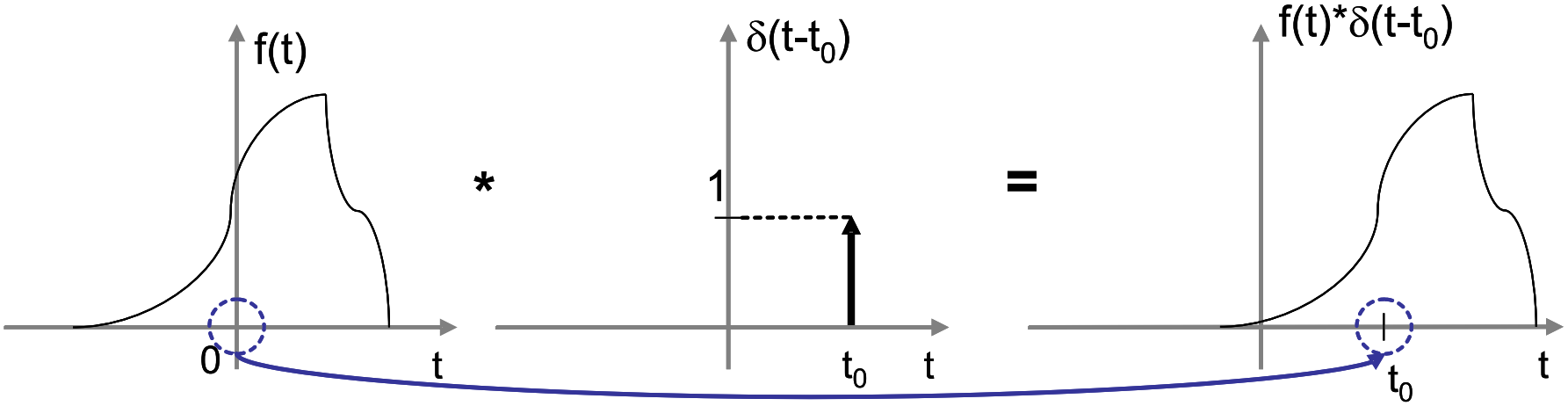
This implies that:

$$\begin{aligned} x(t) * h(t) &= -\int_{+\infty}^{-\infty} x(t-\tau')h(\tau')\mathrm{d}\tau' \\ &= \int_{-\infty}^{+\infty} h(\tau')x(t-\tau')\mathrm{d}\tau' = h(t) * x(t) \end{aligned}$$

Convolution (3)

- ***Proposition:***

$$f(t) * \delta(t - t_0) = f(t - t_0)$$



Convolution (4)

Proof:

$$
\begin{aligned}
f(t) * \delta(t-t_0) &= \int_{-\infty}^{+\infty} f(\tau)\delta(t-t_0-\tau)\mathrm{d}\tau \\
&= \int_{-\infty}^{+\infty} f(\tau)\underbrace{\delta\left(-(\tau-(t-t_0))\right)}_{=\frac{1}{|-1|}\delta(\tau-(t-t_0))}\mathrm{d}\tau \\
&= \int_{-\infty}^{+\infty} f(\tau)\delta\left(\tau-(t-t_0)\right)\mathrm{d}\tau \\
&= \int_{-\infty}^{+\infty} f(t-t_0)\delta(\tau-(t-t_0))\mathrm{d}\tau \\
&= f(t-t_0)\underbrace{\int_{-\infty}^{+\infty}\delta(\tau-(t-t_0))\mathrm{d}\tau}_{=1} = f(t-t_0)
\end{aligned}
$$

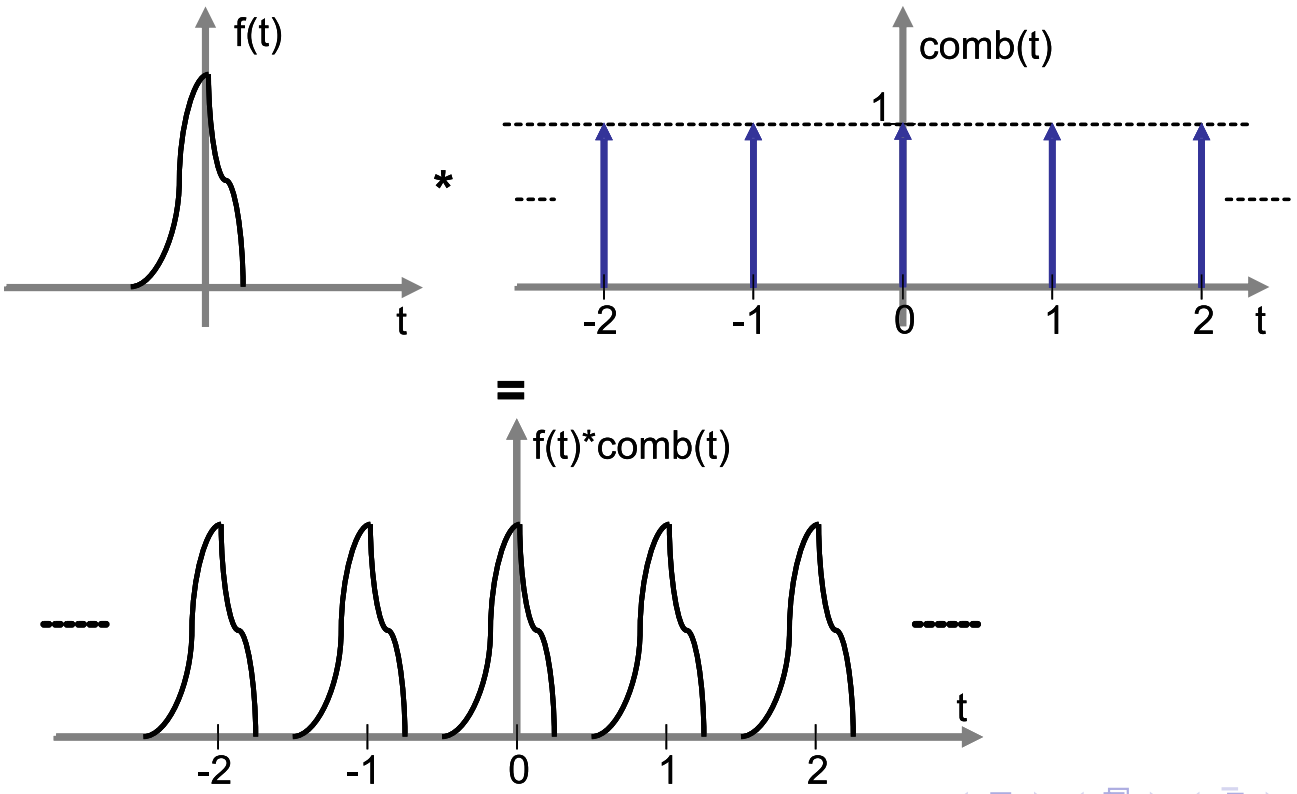
In particular: $f(t) * \delta(t) = f(t)$.

Convolution (5)

- Convolution with the comb function:

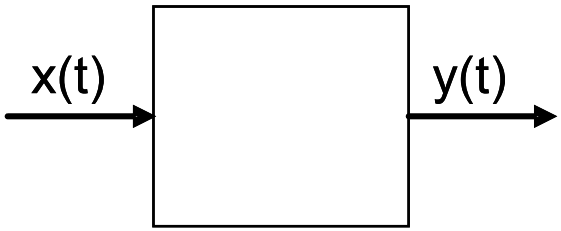
$$f(t) * comb(t) = f(t) * \sum_{n\in\mathbb{Z}} \delta(t-n) = \sum_{n\in\mathbb{Z}} \underbrace{f(t) * \delta(t-n)}_{=f(t-n)}$$

$$= \sum_{n\in\mathbb{Z}} f(t-n)$$



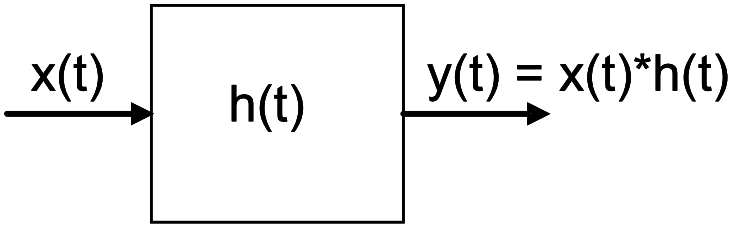
Impulse Response (1)

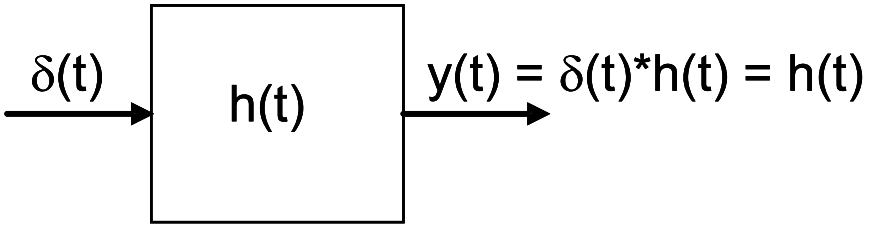
- Consider a time-invariant and linear system:



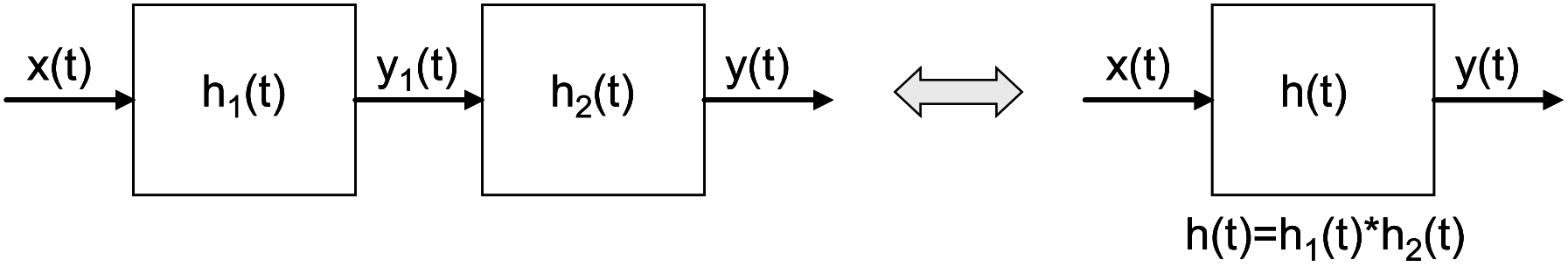
- For an input that is equal to the unit impulse ($x(t) = \delta(t)$), denote by $y(t) = h(t)$ the output of the system.
- $h(t)$ is called the **impulse response** of the system.
- The output of the system can be expressed as:

$$y(t) = x(t) * h(t)$$



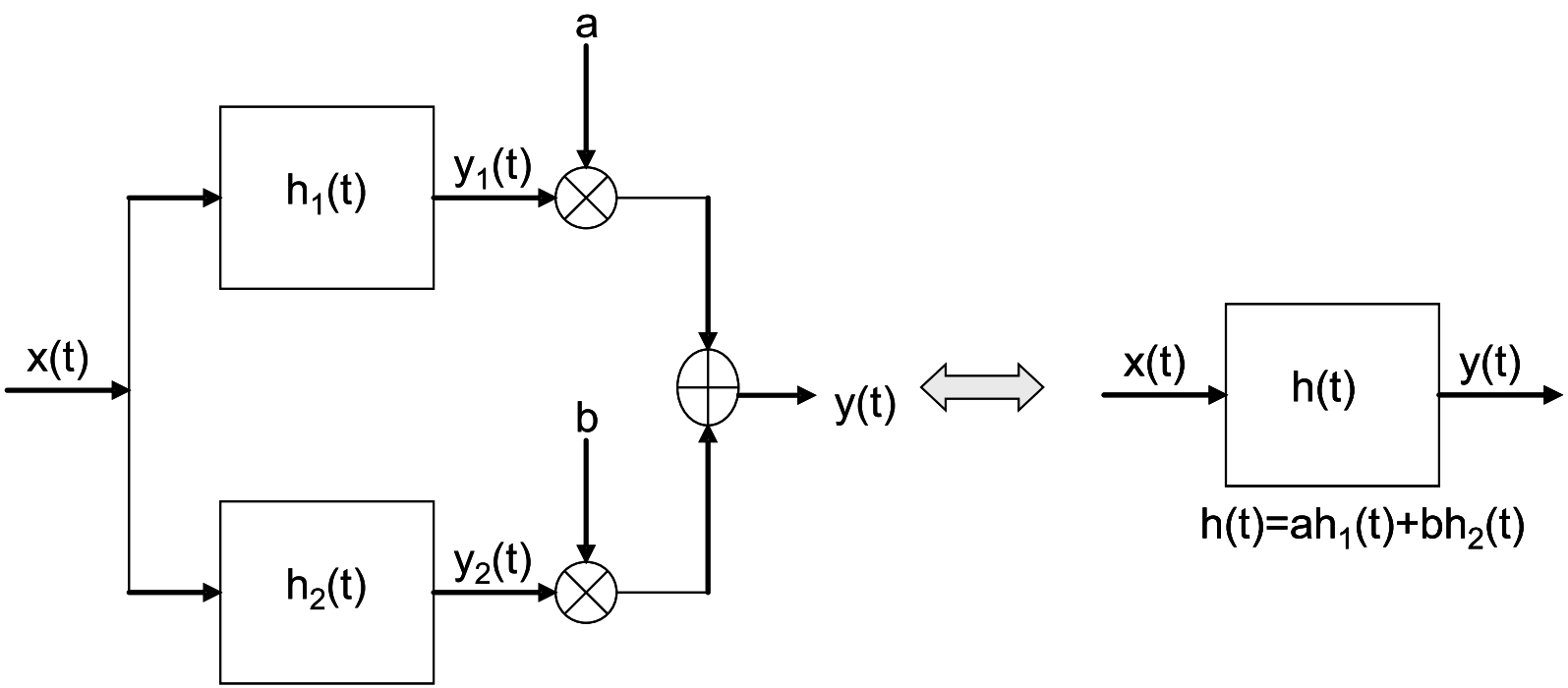


Impulse Response (2)



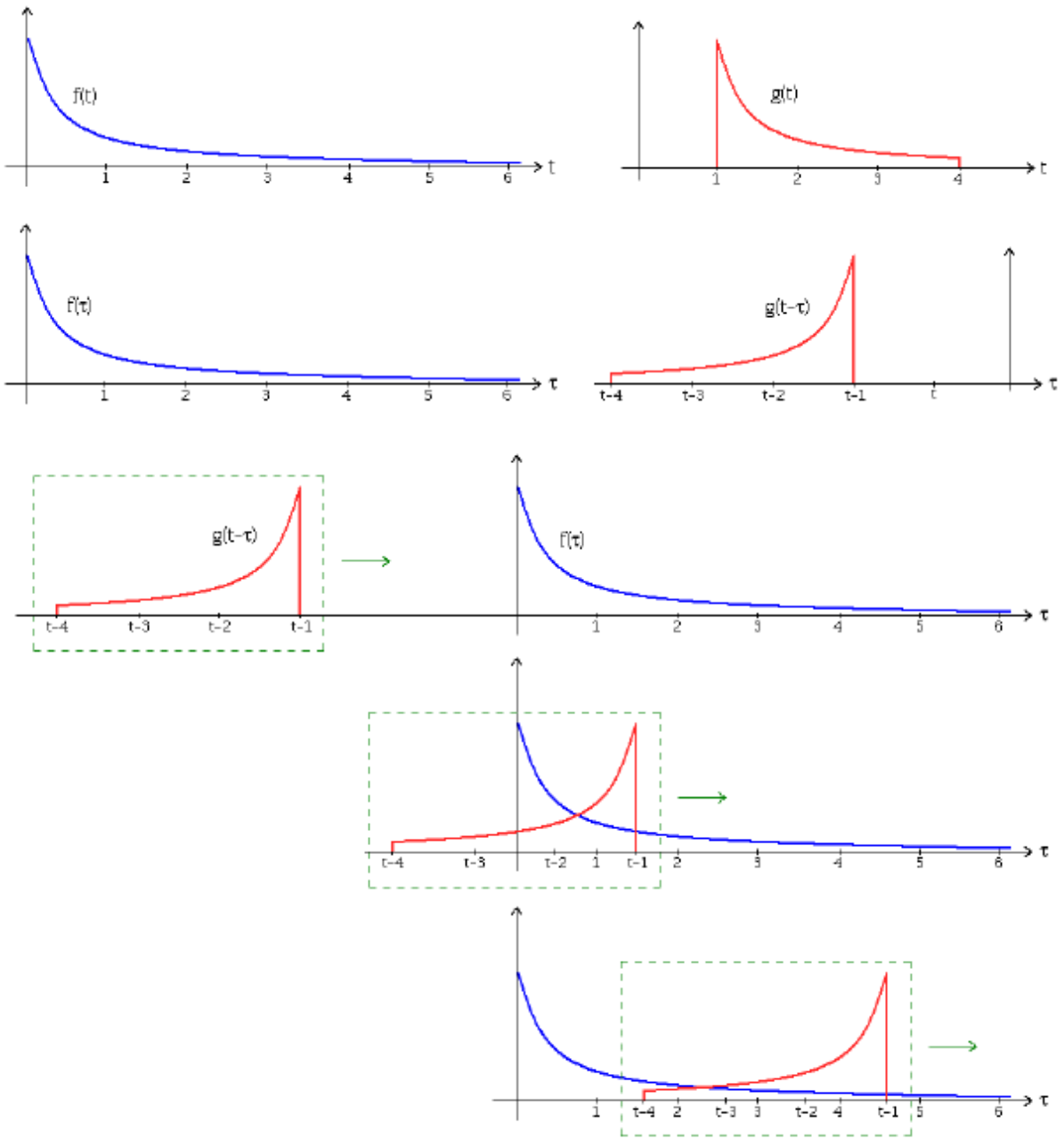
$$
\begin{aligned}
y(t) &= y_1(t) * h_2(t) \\
&= (x(t) * h_1(t)) * h_2(t) \\
&= x(t) * \underbrace{(h_1(t) * h_2(t))}_{h(t)}
\end{aligned}
$$

Impulse Response (3)



$$
\begin{aligned}
y(t) &= ay_1(t) + by_2(t) \\
&= a\left(x(t) * h_1(t)\right) + b\left(x(t) * h_2(t)\right) \\
&= x(t) * (ah_1(t)) + x(t) * (bh_2(t)) \\
&= x(t) * \underbrace{(ah_1(t) + bh_2(t))}_{h(t)}
\end{aligned}
$$

Convolution



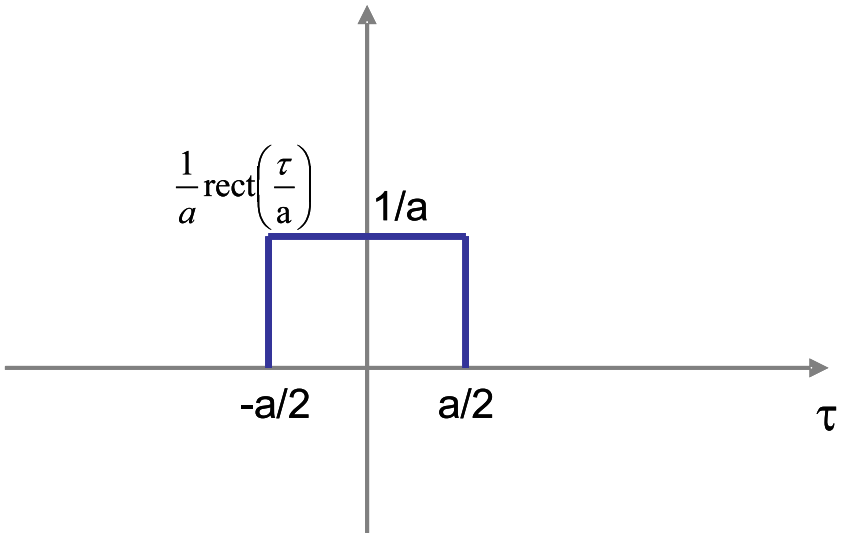
Convolution-Example 1 (1)

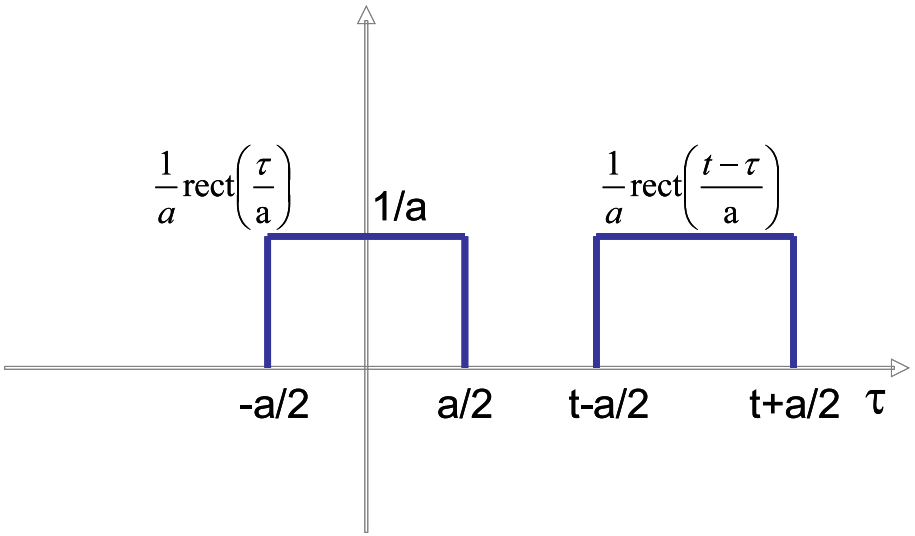
Example 1: Determine the value of the following convolution:

$$f(t) = \frac{1}{a}\text{rect}\left(\frac{t}{a}\right) * \frac{1}{a}\text{rect}\left(\frac{t}{a}\right)$$

- The above convolution can be written as:

$$f(t) = \int_{-\infty}^{+\infty} \frac{1}{a}\text{rect}\left(\frac{\tau}{a}\right) . \frac{1}{a}\text{rect}\left(\frac{t-\tau}{a}\right) \mathrm{d}\tau$$

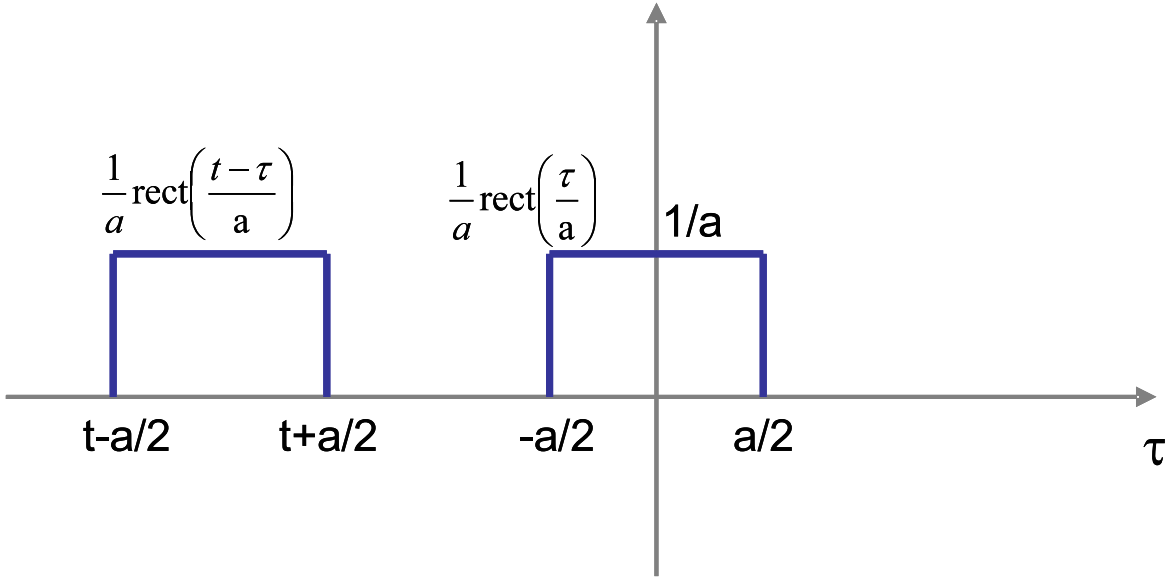




Convolution-Example 1 (2)

Case 1:

$$t + \frac{a}{2} \leq -\frac{a}{2} \Rightarrow t \leq -a$$



In this case:

$$\frac{1}{a}\text{rect}\left(\frac{\tau}{a}\right).\frac{1}{a}\text{rect}\left(\frac{t-\tau}{a}\right) = 0$$

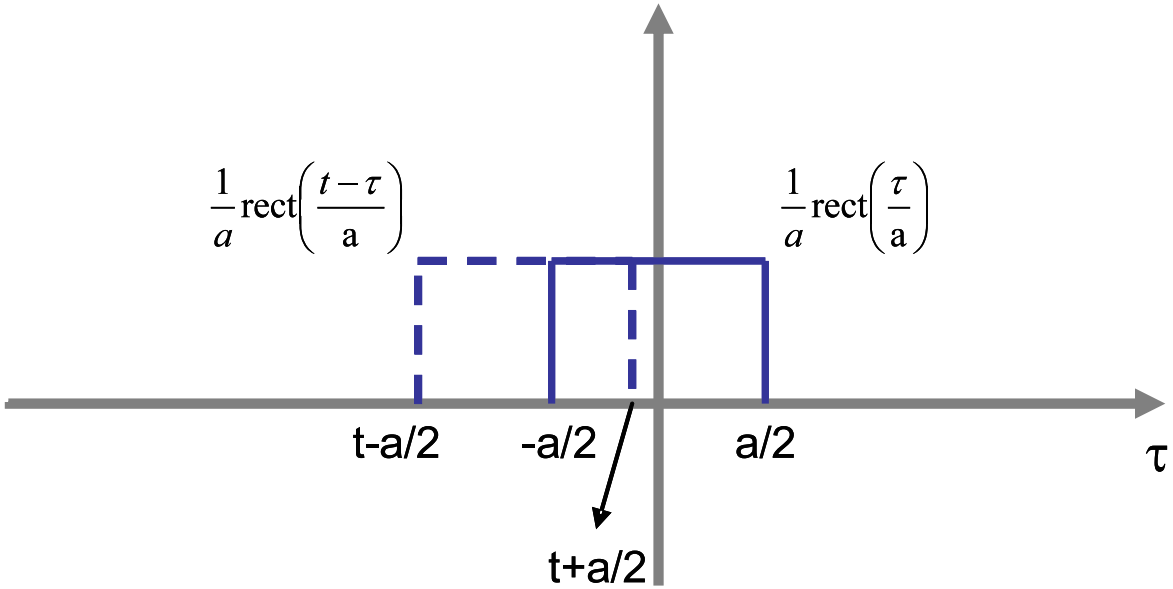
and

$$f(t) = 0$$

Convolution-Example 1 (3)

Case 2:

$$t - \frac{a}{2} \leq -\frac{a}{2} \leq t + \frac{a}{2} \Rightarrow \quad -a \leq t \leq 0$$



In this case:

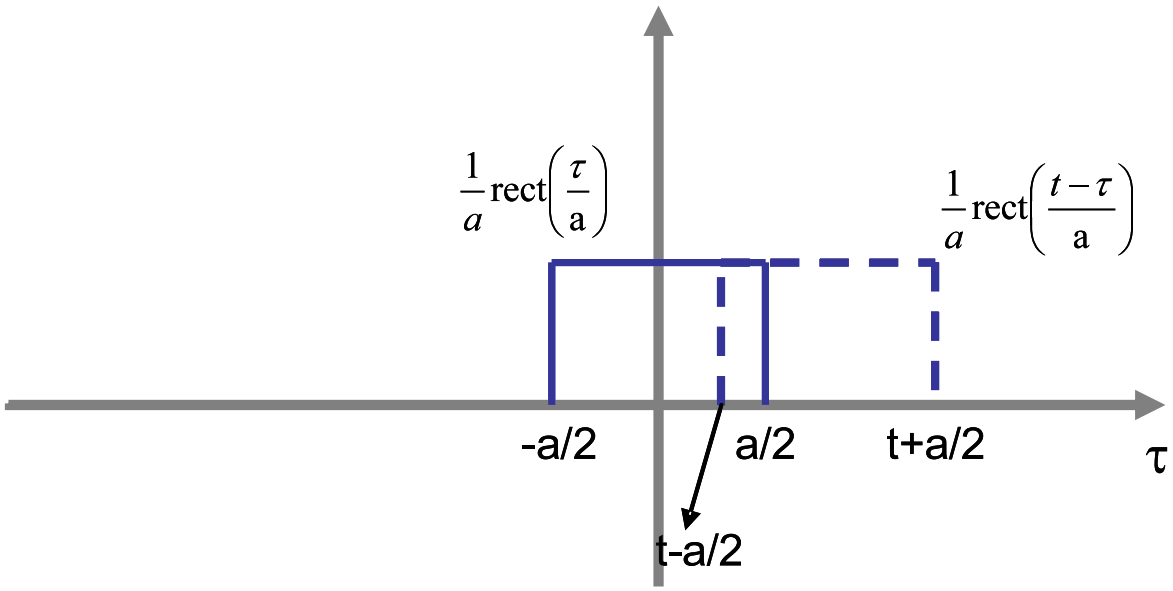
$$f(t) = \int_{-\frac{a}{2}}^{t+\frac{a}{2}} \left(\frac{1}{a}\right)\left(\frac{1}{a}\right) \mathrm{d}\tau = \frac{1}{a^2}\int_{-\frac{a}{2}}^{t+\frac{a}{2}} \mathrm{d}\tau$$

$$= \frac{1}{a^2}\left[\tau\right]_{-\frac{a}{2}}^{t+\frac{a}{2}} = \frac{1}{a^2}(t+a)$$

Convolution-Example 1 (4)

Case 3:

$$t - \frac{a}{2} \leq \frac{a}{2} \leq t + \frac{a}{2} \;\Rightarrow\; 0 \leq t \leq a$$



In this case:

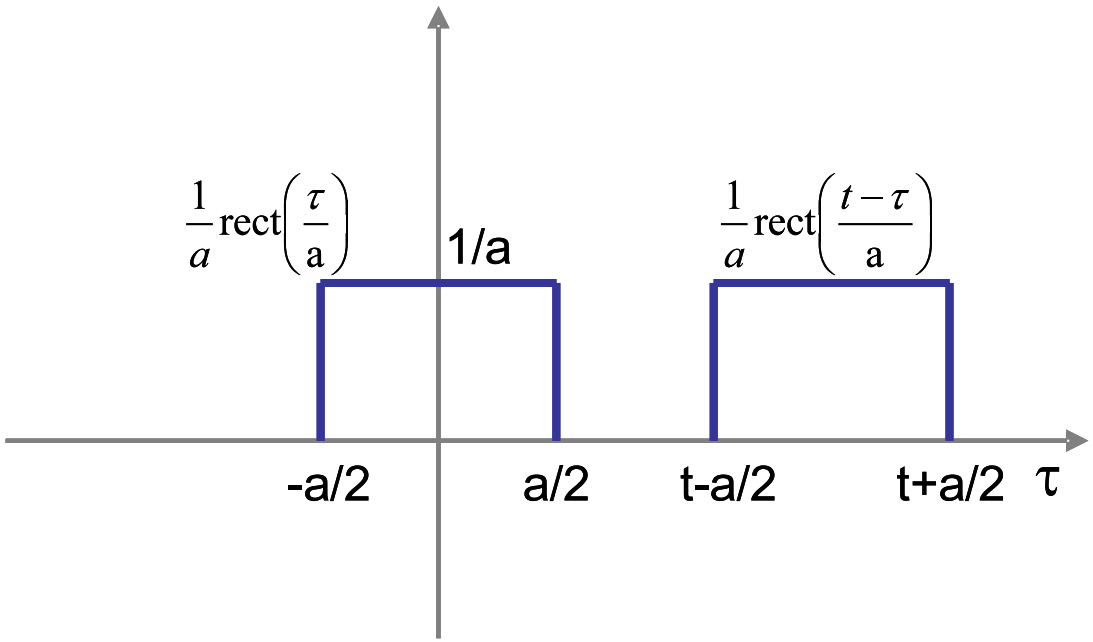
$$f(t) = \int_{t-\frac{a}{2}}^{\frac{a}{2}} \left(\frac{1}{a}\right)\left(\frac{1}{a}\right) \mathrm{d}\tau = \frac{1}{a^2}\int_{t-\frac{a}{2}}^{\frac{a}{2}} \mathrm{d}\tau$$

$$= \frac{1}{a^2}\left[\tau\right]_{t-\frac{a}{2}}^{\frac{a}{2}} = \frac{1}{a^2}(-t+a)$$

Convolution-Example 1 (5)

Case 4:

$$t - \frac{a}{2} \geq \frac{a}{2} \Rightarrow t \geq a$$



$$\frac{1}{a}\text{rect}\left(\frac{\tau}{a}\right).\frac{1}{a}\text{rect}\left(\frac{t-\tau}{a}\right) = 0$$

and

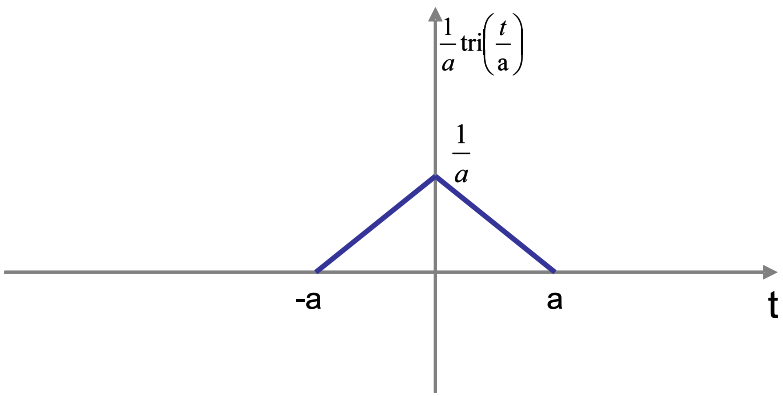
$$f(t) = 0$$

Convolution-Example 1 (6)

- Consequently:

$$f(t) = \begin{cases} 0 & \text{for} \quad t \leq -a \\ \frac{1}{a^2}(t+a) & \text{for} \quad -a \leq t \leq 0 \\ \frac{1}{a^2}(-t+a) & \text{for} \quad 0 \leq t \leq a \\ 0 & \text{for} \quad t \geq a \end{cases}$$

- Plotting these curves, we obtain:

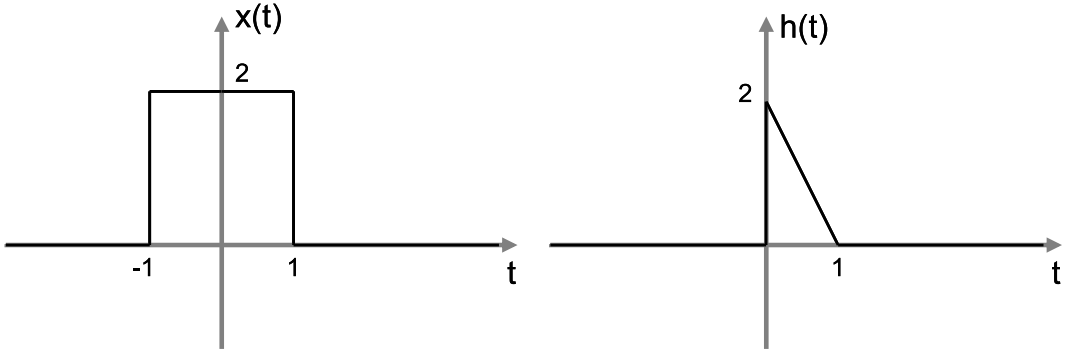


- This implies that:

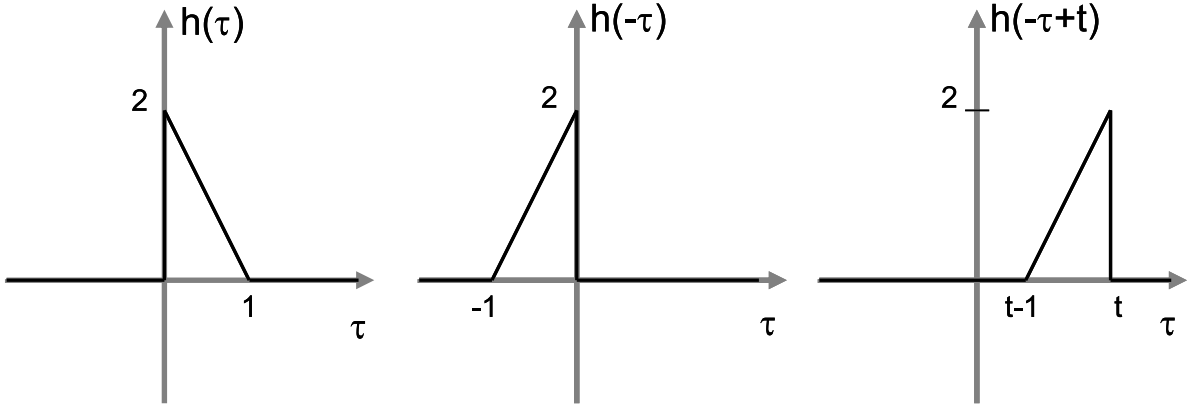
$$\frac{1}{a}\text{rect}\left(\frac{t}{a}\right) * \frac{1}{a}\text{rect}\left(\frac{t}{a}\right) = \frac{1}{a}\text{tri}\left(\frac{t}{a}\right)$$

Convolution-Example 2 (1)

Example 2: Determine the convolution of the two functions $x(t)$ and $h(t)$ given by:



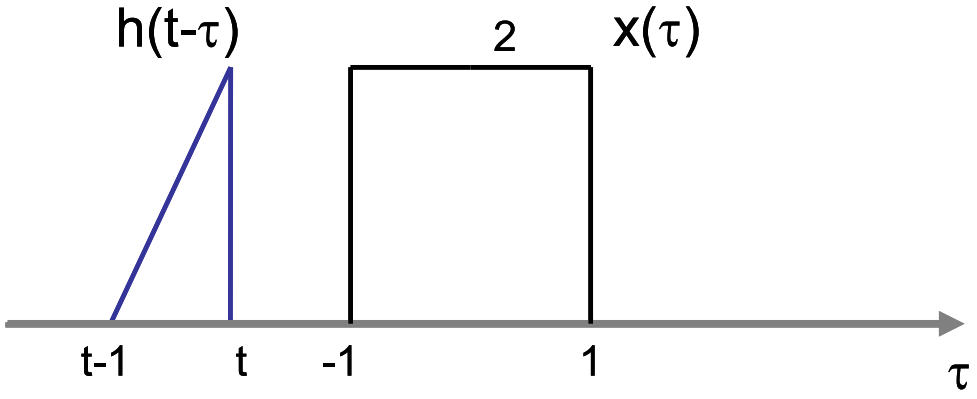
- $h(t-\tau)$ can be plotted (as a function of τ) as follows:



Convolution-Example 2 (2)

Case 1:

$$t \leq -1$$

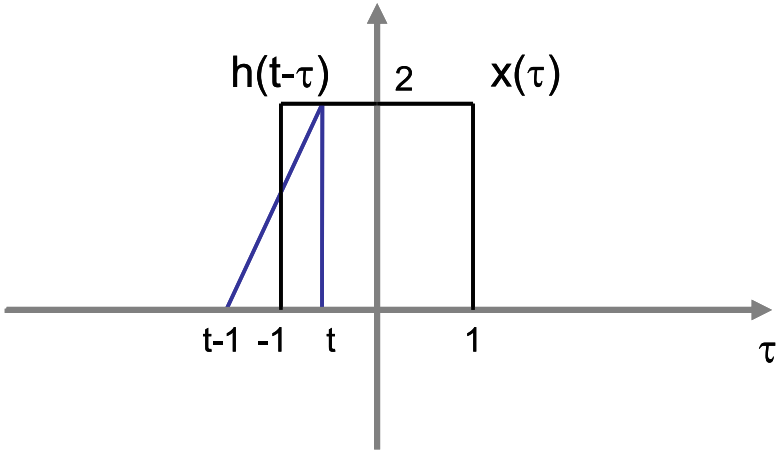


$$x(\tau)h(t-\tau) = 0 \quad \Rightarrow \quad f(t) = \int_{-\infty}^{+\infty} x(\tau)h(t-\tau)\mathrm{d}\tau = 0$$

Convolution-Example 2 (3)

Case 2:

$$t-1 \leq -1 \leq t \quad \Rightarrow \quad -1 \leq t \leq 0$$



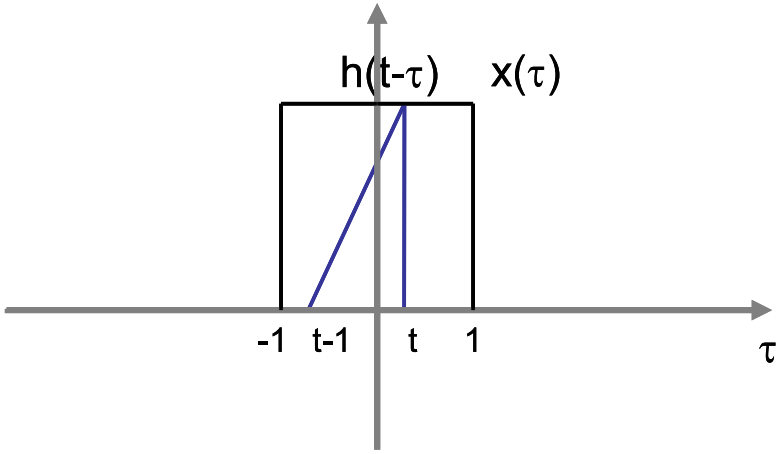
$$f(t) = \int_{-1}^{t} 2[\underbrace{2(\tau-(t-1))}_{h(t-\tau)}]\mathrm{d}\tau = 4\left[\frac{\tau^2}{2} - (t-1)\tau\right]_{-1}^{t}$$

$$= 4\left[\frac{t^2}{2} - t^2 + t - \left[\frac{1}{2} + t - 1\right]\right] = 2(1-t^2)$$

Convolution-Example 2 (4)

Case 3:

$$t-1 \geq -1 \;\&\; t \leq 1 \quad \Rightarrow \quad 0 \leq t \leq 1$$



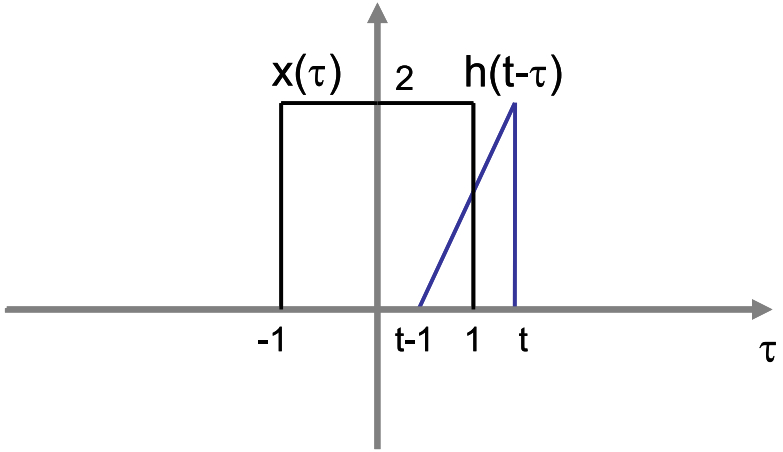
$$f(t) = \int_{t-1}^{t} 2[\underbrace{2(\tau-(t-1))}_{h(t-\tau)}]\mathrm{d}\tau = 4\left[\frac{\tau^2}{2} - (t-1)\tau\right]_{t-1}^{t}$$

$$= 4\left[\frac{t^2}{2} - t^2 + t - \left[\frac{(t-1)^2}{2} - (t-1)^2\right]\right] = 2$$

Convolution-Example 2 (5)

Case 4:

$$t-1 \le 1 \le t \quad \Rightarrow \quad 1 \le t \le 2$$



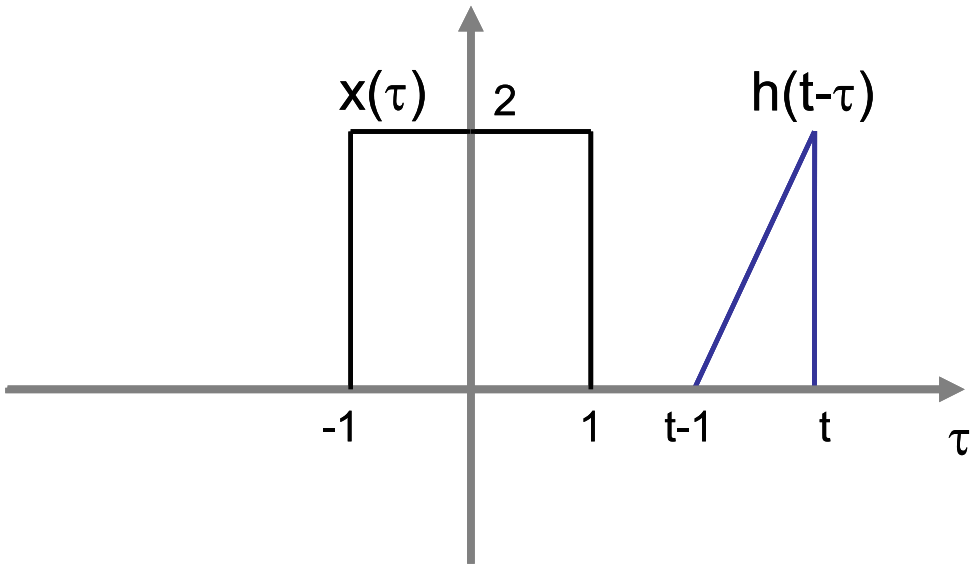
$$f(t) = \int_{t-1}^{1} 2[\underbrace{2(\tau-(t-1))}_{h(t-\tau)}]\mathrm{d}\tau = 4\left[\frac{\tau^2}{2} - (t-1)\tau\right]_{t-1}^{1}$$

$$= 4\left[\frac{1}{2} - t + 1 - \left[\frac{(t-1)^2}{2} - (t-1)^2\right]\right] = 2(t-2)^2$$

Convolution-Example 2 (6)

Case 5:

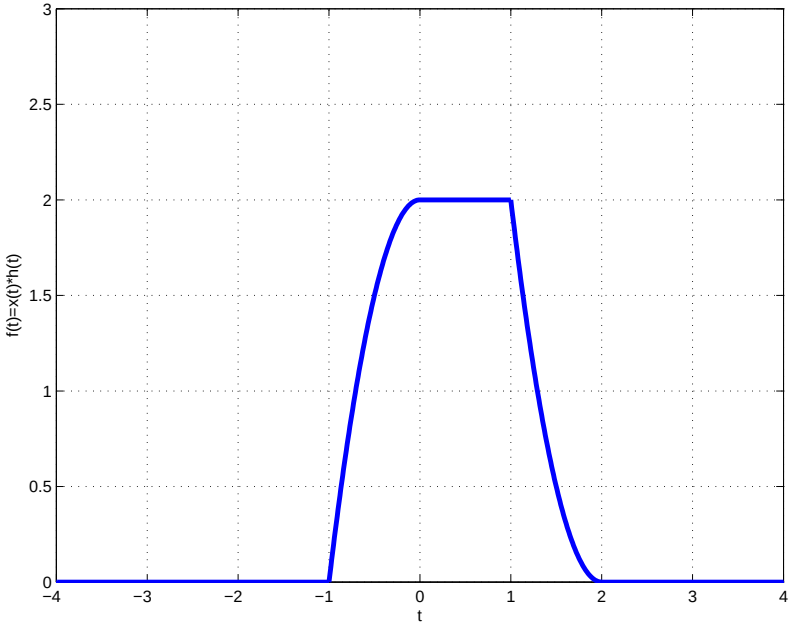
$$t - 1 \geq 1 \quad \Rightarrow \quad t \geq 2$$



$$x(\tau)h(t-\tau) = 0 \quad \Rightarrow \quad f(t) = \int_{-\infty}^{+\infty} x(\tau)h(t-\tau)\mathrm{d}\tau = 0$$

Convolution-Example 2 (7)

- Consequently:

$$f(t) = \begin{cases} 0 & \text{for} \quad t \leq -1 \\ 2(1-t^2) & \text{for} \quad -1 \leq t \leq 0 \\ 2 & \text{for} \quad 0 \leq t \leq 1 \\ 2(t-2)^2 & \text{for} \quad 1 \leq t \leq 2 \\ 0 & \text{for} \quad t \geq 2 \end{cases}$$

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